

# BUDGET SETS, PREFERENCES, AND CHOICE PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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## Problem 1

Violet is preparing for exams in economics and sociology. She has time to study 40 pages of economics and 30 pages of sociology. In the same amount of time she could also read 30 pages of economics and 60 pages of sociology.

- Assuming that the number of pages per hour that she can read of either subject does not depend on how she allocates her time, how many pages of sociology could she read if she decided to spend all of her time on sociology and none on economics? *Hint: you have two points on her budget line, so you should be able to determine the entire line.*
- How many pages of economics could she read if she decided to spend all of her time reading economics?
- Draw Violet's budget line. Be sure to label all relevant features (intercepts and slope).
- Suppose now that Violet has a friend, Brenda, who took in the previous term the same Economics class that Violet is taking. However, from the topics that Violet has to study, Brenda only remembers the first few chapters. Thus, Brenda can only help with the first 25 pages that Violet has to study. For these first 25 pages, with the help of Brenda, Violet has time to study 10 pages of economics and 140 pages of sociology, whereas in the same amount of time, she could also read 20 pages of economics and 130 pages of sociology. How many pages of economics can Violet now read if she decides to spend all of her time reading economics? How many of Sociology?
- Draw Violet's new budget line after she receives help from Brenda. Be sure to label all relevant features, like intercepts slopes or *kinks*.

## Problem 2

On the planet of Mongo, emperor *Ming the Merciless* forces his subjects to go through a rather contrived market exchange procedure: Mongo has two kinds of currencies, **blue money** and **red money**. Every commodity has two prices—a **red-money** price and a **blue-money** price. Every Mongoan has two incomes—a **red** income and a **blue** income.

In order to buy an object, a Mongoan has to pay that object's **red-money** price in **red** money and its **blue-money** price in **blue** money. It is forbidden to trade one kind of money for the other, and this prohibition is strictly enforced by *Ming's* ruthless enforcers.

- There are just two consumer goods on Mongo, ambrosia and fuel. All Mongoans prefer more of each good to less.

- The **blue** prices are **1 bcu** (bcu stands for blue currency unit.) per unit of ambrosia and **1 bcu** per unit of fuel.
  - The **red** prices are **2 rcus** (red currency units) per unit of ambrosia and **6 rcus** per unit of fuel.
- (a) Draw the **red budget** (with red ink) and the **blue budget** (with blue ink) for a Mongoan named Barin whose **blue income is 10** and his **red income is 30**. Shade in the “budget set” containing all of the commodity bundles that Barin can afford, given his two budget constraints. Remember, Barin has to have enough blue money and enough red money to pay both the blue-money cost and the red-money cost of a bundle of goods.
- (b) Another Mongoan, Aura, faces the same prices that Barin faces and has the **same red income** as Barin, but Aura has a **blue income of 20**. Explain how it is that Aura will not spend its entire blue income no matter what her tastes may be. *Hint: Draw Aura’s budget lines.*
- (c) A group of radical economic reformers arrive to Mongo and believe that the currency rules are unfair. “Why should everyone have to pay two prices for everything?” they ask. They propose the following scheme. Mongo will continue to have two currencies, every good will have a blue price and a red price, and every Mongoan will have a blue income and a red income. But nobody has to pay two prices. Instead, everyone on Mongo must declare itself to be either a **Blue-Money Purchaser** (a “Blue”) or a **Red-Money Purchaser** (a “Red”) before he or she buys anything at all. **Blues** must make all their purchases in blue money at the blue prices, spending only their blue incomes. **Reds** must make all of their purchases in red money, spending only their red incomes. Suppose that Barin has the same income after this reform and that prices do not change. Before declaring which kind of purchaser he will be, Barin contemplates the set of commodity bundles that he could afford by making one declaration or the other. Let us call a commodity bundle “attainable” if Barin can afford it by declaring himself to be a **Blue** and buying the bundle with blue money *or* if Barin can afford the bundle by declaring itself to be a **Red** and buying it with red money. Draw correctly labeled diagram in which you shade in all of the attainable bundles.
- (d) Are Mongoan budgets really so fanciful? Can you think of situations on Earth where people must simultaneously satisfy more than one budget constraint? Is money the only scarce resource that people use up?

### Problem 3

In this exercise, you will explore a preference relation and the utility that represents it that incorporates a *satiation point*. A satiation point is a bundle that provides complete “bliss” for the consumer. Suppose the bliss point is  $(b_1^*, b_2^*)$ . The ranking of all other bundles is determined by how close they are to  $(x^*, y^*)$ . Thus we have

$$(x_1, x_2) \succeq (y_1, y_2) \Leftrightarrow \sqrt{(x_1 - b_1^*)^2 + (x_2 - b_2^*)^2} \leq \sqrt{(y_1 - b_1^*)^2 + (y_2 - b_2^*)^2}$$

- (a) Strawberry Shortcake likes to spend some time studying and some time dating. Her preferences can be represented by the aforementioned utility function, whose indifference curves between hours per week spent studying and hours per week spent dating are concentric circles around her favorite combination, which is 20 hours of studying and 15 hours of dating per week. Surely, the closer she is to her favorite combination, the happier she is.

- (b) Graph some indifference curves for this preference relation. (Hint: A circle with center  $(b_1^*, b_2^*)$  and radius  $k$  is defined by the equation  $(x_1 - b_1^*)^2 + (x_2 - b_2^*)^2 = k^2$ . You may also want to use the monotonic transformation  $f(u) = u^2$  defined for any positive real number  $u$ .)
- (c) Suppose that Strawberry is currently studying 25 hours a week and dating 3 hours a week. Would she prefer to be studying 30 hours a week and dating 8 hours a week? Justify your answer.
- (d) The MRS of good 1 versus good 2 for this preference relation is given by

$$MRS = \frac{x_1 - b_1}{x_2 - b_2}.$$

Compute the slope of the indifference curves at the combinations of studying 25 hours a week/dating 3 hours a week and 30 hours a week/dating 8 hours a week.

## Problem 4

Bert has preferences represented by the utility function  $u(a, b) = ab/100$ , where  $a$  is the number of ounces of animal crackers that he consumes and  $b$  is the number of ounces of broccoli that he consumes.

- (a) Sketch the set of points that Bert finds indifferent to having 8 ounces of animal crackers and 2 ounces of broccoli. Also sketch the locus of points that Bert finds indifferent to having 6 ounces of animal crackers and 4 ounces of broccoli.
- (b) Ernie has preferences represented by the utility function  $v(a, b) = 1000a^2b^2$ , where  $a$  is the number of ounces of animal crackers that he consumes and  $b$  is the number of ounces of broccoli that he consumes. Sketch the set of points that Ernie finds indifferent to having 8 ounces of animal crackers and 2 ounces of broccoli. Also, sketch the locus of points that Ernie finds indifferent to having 6 ounces of animal crackers and 4 ounces of broccoli.
- (c) Are Bert's preferences convex? Are Ernie's?
- (d) What can you say about the difference between the indifference curves you drew for Bert and those you drew for Ernie?
- (e) How could you tell this was going to happen without having to draw the curves? Justify your answer.

## Problem 5

Zarina wants to buy a new car and has a budget of \$25,000. She has just found a magazine that assigns each car an index for styling and an index for gas mileage. Each index runs from 1 to 10, with 10 representing either the most styling or the best gas mileage. While looking at the list of cars, Zarina observes that on average, as the style index increases by one unit, the price of the car increases by \$5,000. She also observes that as the gas-mileage index rises by one unit, the price of the car increases by \$2,500.

- (a) Assuming you put gas mileage on the horizontal axis, what is the equation for Zarina's budget constraint?

- (b) Suppose that Zarina's preferences are such that she always receives three times as much satisfaction from an extra unit of styling, as she does from gas mileage, i.e., her preferences can be represented as  $U(S, G) = 3S + G$ , where  $S$  are units of the styling index and  $G$  are units of the gas mileage index. What type of car will Zarina choose?
- (c) Suppose that Zarina's marginal rate of substitution (of gas mileage for styling) is equal to  $S/4G$ . What value of each index would she like to have in her car?
- (d) Suppose that Zarina's marginal rate of substitution (of gas mileage for styling) is equal to  $3S/G$ . What value of each index would she like to have in her car?

## Problem 6

Dastan runs a restaurant. One of the dishes in the menu is a meat and potato stew. Dastan has a budget of \$200 that he allocates to the purchase of the main ingredients: meat and potatoes. Suppose meat costs \$4 per pound and potatoes cost \$2 per pound.

Suppose Dastan's utility function for meat and potatoes is given by  $U(P, M) = P^{1/4}M^{3/4}$ . The marginal rate of substitution (of potatoes for meat) of this utility function is  $\frac{M}{3P}$ .

- (a) Dastan buys his ingredients at a wholesaler supermarket that has the following special promotion. If he buys 20 pounds of potatoes (at \$2 per pound), he gets the next 10 pounds for free. This offer applies only to the first 20 pounds she buys. All potatoes in excess of the first 20 pounds (excluding bonus potatoes) are still \$2 per pound. *Hint: It is a good idea that you draw Dastan's budget constraint.* What is the combination of meat and potatoes that maximizes Dastan's utility?
- (b) Now suppose that there is an outbreak of *Pectobacterium carotovorum* (this bacteria causes Potato Rot) that raises the price of potatoes to \$4 per pound. The wholesaler supermarket ends the promotion. What is the combination of meat and potatoes that maximizes Dastan's utility?

## Problem 7

Mr. Bond derives utility from vodka martinis ( $m$ ) in proportion to the number he drinks:

$$U(m) = m.$$

However, Mr. Bond is very particular about his martinis: He only enjoys them made in the exact proportion of two parts vodka ( $b$ ) to one part vermouth ( $v$ ), and of course, shaken, not stirred. Hence, we can rewrite Mr. Bond's utility function as

$$U(m) = U(b, v) = \min\left(\frac{b}{2}, v\right).$$

- (a) Suppose that in a diagram, you put vodka on the horizontal axis and vermouth on the vertical axis. The equation that describes the points at which the kinks of Mr. Bond's indifference curves are is given by the expression: (Write your answer in decimal format, not as a fraction)
- (b) Suppose that the price of vermouth is \$2 per ounce and the price of vodka is \$6 per ounce. Mr. Bond has an income of \$140. Mr. Bond will buy:

- (c) A regular cocktail glass has a volume of 3 ounces. How many martinis will Mr. Bond drink?
- (d) What is the cost per martini?